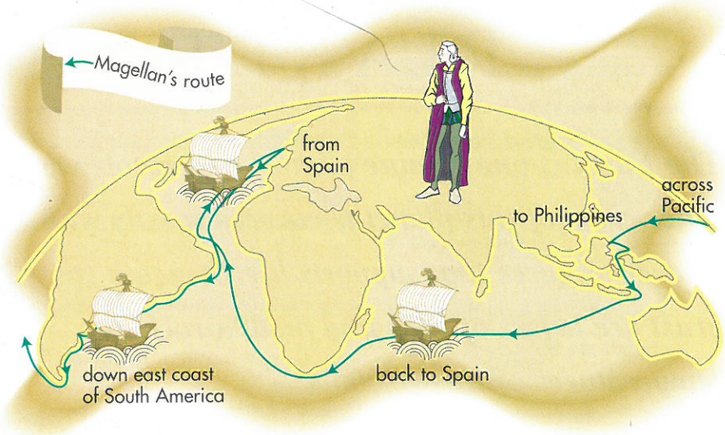
# 6.1 Earth in the round

Until the sixteenth century, most people thought that the Earth was flat. There was plenty of evidence to support that belief. Many sailing ships did not return from long journeys and it was assumed that they simply fell over the edge of the Earth.

When Ferdinand Magellan sailed west from Spain in 1519 to reach the Spice Islands (now the Moluccas, part of Indonesia), known at the time to be east of Spain, he was ridiculed.

However, one of his ships, the *Victoria*, found the Spice Islands and continued westward, eventually arriving home. Magellan died during the journey, but the *Victoria* was the first ship to sail around the world. The journey provided strong evidence that the Earth was round.

Photographs taken of Earth from space show that it is almost spherical in shape, being very slightly flattened at the north and south poles. The **equator** is an imaginary circle around the Earth that divides it into two halves, called the northern hemisphere and southern hemisphere.




Magellan's journey provided strong evidence that the Earth was round.

## Structure of the Earth

### Crust

The outer layer or **crust** of the Earth is made mostly of solid rock. It is a bit like a shell and covers the whole planet. It varies in thickness from about 8 km below the oceans to about 40 km below the continents.

### Mantle

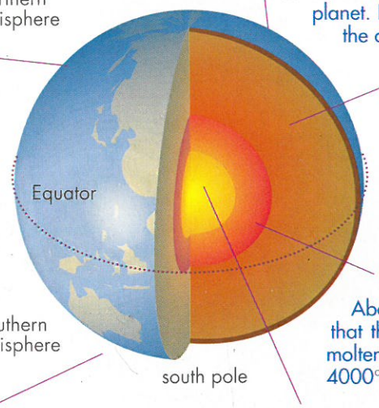
Below the crust is a region of partially molten rock. This **mantle** is about 2900 km thick and its temperature varies between about 500°C near the crust to about 2200°C at its deepest part.

### Outer core

About 2300 km thick, the **outer core** is so hot that the iron and nickel from which it is made is molten. The temperature varies from about 4000°C to about 6000°C.

### Inner core

Even though the temperature of the **inner core** may be as high as 7000°C, the iron and nickel in this part of the Earth is believed to be solid. This is because of all the weight pushing down on it. The inner core is about 1200 km thick.




### At the surface

Only about 30 per cent of the surface of the Earth is dry land. The rest of the surface is water. About 97 per cent of the water at the surface is the salt water that makes up the oceans and seas. About 2 per cent of the water is stored as ice near the north and south poles. The rest of the water at the surface is the fresh water that can be found in rivers, streams, lakes and ponds.

### Above the surface

The region of gases above the surface of the Earth is called the **atmosphere**. It is a mixture of gases called **air**. The air in the atmosphere is essential for life on the planet. About 99 per cent of the air in the atmosphere is less than 80 km above the surface.



# Activities

## Using data

The centre of the Earth is about 6370 km from the surface. In 1864, the science fiction author Jules Verne wrote the novel *Journey to the Centre of the Earth*. It tells of an amazing journey through the inside of the Earth that begins with a descent into a volcano in Iceland. In fact, the deepest mines go down only 4 km into the Earth. The deepest drill hole into the Earth is about 14 km deep. The following table shows the temperature measured at different depths in a drill hole.

**Temperature at different depths of a drill hole**

| Depth (km) | Temperature (°C) |
|---|---|
| 0 | 15 |
| 1 | 44 |
| 2 | 73 |
| 3 | 102 |
| 4 | 130 |
| 5 | 158 |
| 6 | 187 |
| 7 | 215 |
| 8 | 242 |

1. Plot a graph to show how the temperature increases with depth. You will need to label the horizontal axis 'depth (km)' and the vertical axis 'temperature (°C)'.
2. Use your graph to predict the temperature at the following depths:
   (a) 2.5 km
   (b) 10 km.
3. About how many degrees does the temperature increase for each kilometre below the surface?
4. Use your answer to question 3 to predict the temperature at the centre of the Earth. Scientists estimate the temperature at the centre of the Earth to be about 7000°C. Explain why use of the data in question 3 gives such a high prediction.

## Remember

1. Name the four major regions below the Earth's surface.
2. Which of the regions below the Earth's surface are believed to be solid?
3. How much of the Earth's surface is covered with water?

## Think

1. At the time of Magellan's journey in 1519, most people believed that the Earth was flat. Was there any other evidence at that time to suggest that the Earth was round? What was it?
2. Members of the Flat Earth Society believe that the Earth is flat and shaped like a dinner plate. They believe that photographs taken from space that show the Earth to be a sphere are part of a giant hoax. What do you think? Write down some observations that support your opinion.
3. Even though the inner core of the Earth is hotter than the molten outer core, it is believed to be solid. How could this be?
4. The Earth is travelling through space at a speed of about 110 000 kilometres per hour. This means that it covers about 30 kilometres every second. How far would the Earth travel in
   (a) a week?
   (b) a year?

## Create

Draw a scale diagram of the Earth using the following instructions. You will need a compass, a pencil and a sheet of plain A4 paper.

- Open up the compass so that the ends are 10 cm apart. Use it carefully to draw a circle with a radius of 10 cm.
- Using the same centre, draw two more circles, one of radius 5.5 cm and one of radius 1.9 cm.
- You have now shown three of the regions below the Earth's surface. Label the three regions.
- On your scale diagram, the crust would need to be represented by a pencil line on the surface. A thick pencil line would represent the thickest part of the crust while a thin pencil line would represent the thinnest part. On your scale diagram, the atmosphere would be about 2 mm thick. A thick blue line could be used to represent the atmosphere.
- Complete the labelling of your diagram and then colour it.